

Atelier 6 Master DSBD & IA

Virtualisation des Réseaux avec SDN, Mininet et le controller OpenDaylight

- Mininet est un simulateur de réseau open-source qui permet de créer des réseaux virtuels pour tester et expérimenter des protocoles réseau, des topologies et des contrôleurs SDN (Software-Defined Networking) dans un environnement de test.
- -Mininet permet de tester différents scénarios de réseau sans avoir besoin de matériel physique.
- Mininet est souvent utilisé pour tester les réseaux SDN tels que OpenDaylight, où un contrôleur gère la logique du réseau, comme le routage et la commutation, plutôt que les commutateurs physiques eux-mêmes.

1. Préparation du système

Avant d'installer tout logiciel requis, assurez-vous que votre système est à jour et que toutes les dépendances sont installées.

Commands:

```
sudo apt update -y
```

2. Installing Java JRE

OpenDaylight nécessite Java pour fonctionner. Nous installerons OpenJDK 8 car il est compatible avec OpenDaylight 0.8.4:

```
sudo apt-get install openjdk-8-jre
```

Vérifier que vous utilisez la bonne version :

```
sudo update-alternatives --config java
```



```
dsbd@OLD-controller: ~  
dsbd@OLD-controller:~$ sudo update-alternatives --config java  
There is only one alternative in link group java (providing /usr/bin/java): /usr  
/lib/jvm/java-8-openjdk-amd64/jre/bin/java  
Nothing to configure.  
dsbd@OLD-controller:~$
```

Ajouter la variable d'environnement JAVA_HOME et vérifier :

```
sudo echo 'export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/jre' >> ~/.bashrc
```

```
source ~/.bashrc
```

```
echo $JAVA_HOME
```

A terminal window titled 'dsbd@OLD-controler: ~' with standard window controls. The terminal shows the following commands and output:

```
dsbd@OLD-controler:~$ sudo echo 'export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/jre' >> ~/.bashrc
dsbd@OLD-controler:~$ source ~/.bashrc
dsbd@OLD-controler:~$ echo $JAVA_HOME
/usr/lib/jvm/java-8-openjdk-amd64/jre
dsbd@OLD-controler:~$
```

3. Downloading OpenDaylight

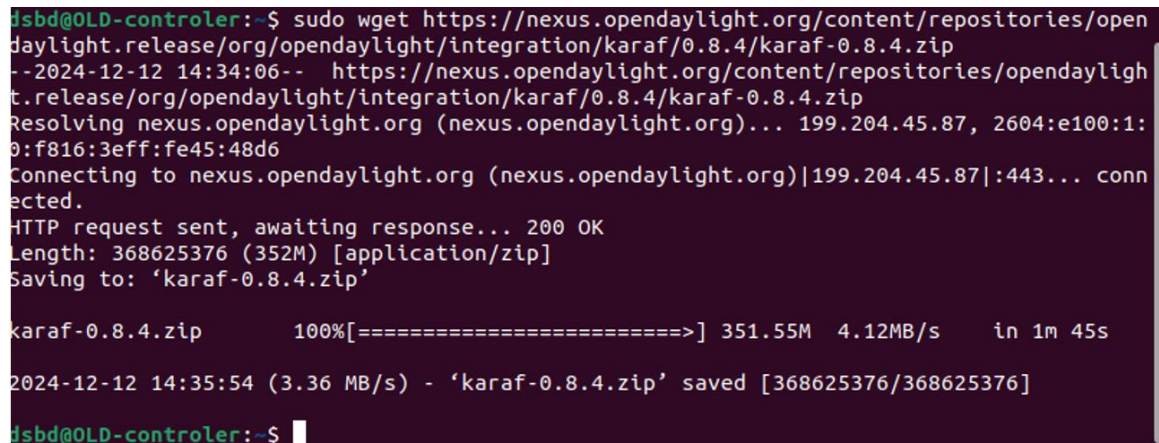
Téléchargez et extraire OpenDaylight. On utilise la version 0.8.4 d'OpenDaylight.

```
sudo apt install wget
```

```
sudo wget https://nexus.opendaylight.org/content/repositories/opendaylight.release/org/opendaylight/integration/karaf/0.8.4/karaf-0.8.4.tar.gz
```

```
sudo install tar
```

```
tar -xf karaf-0.8.4.tar.gz
```

A terminal window showing the execution of the wget command to download a file from a Nexus repository. The output shows the file being resolved, connected to, and downloaded successfully.

```
dsbd@OLD-controler:~$ sudo wget https://nexus.opendaylight.org/content/repositories/opendaylight.release/org/opendaylight/integration/karaf/0.8.4/karaf-0.8.4.zip
--2024-12-12 14:34:06-- https://nexus.opendaylight.org/content/repositories/opendaylight.release/org/opendaylight/integration/karaf/0.8.4/karaf-0.8.4.zip
Resolving nexus.opendaylight.org (nexus.opendaylight.org)... 199.204.45.87, 2604:e100:1:0:f816:3eff:fe45:48d6
Connecting to nexus.opendaylight.org (nexus.opendaylight.org)|199.204.45.87|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 368625376 (352M) [application/zip]
Saving to: 'karaf-0.8.4.zip'

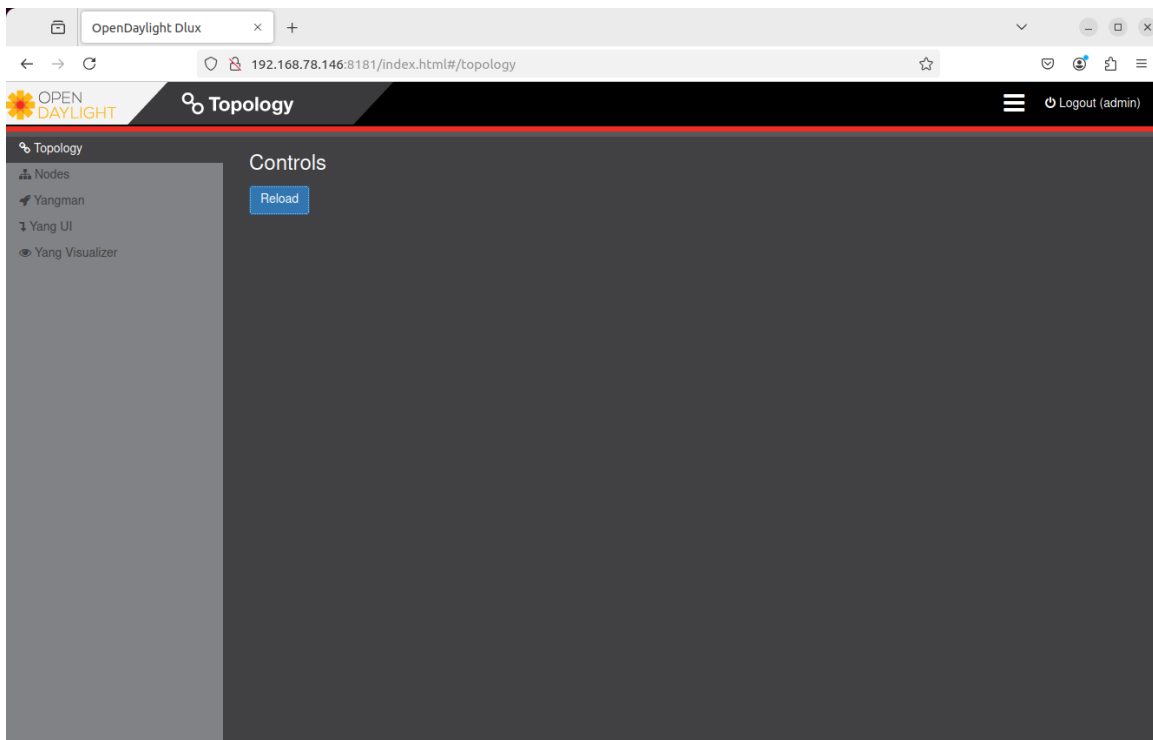
karaf-0.8.4.zip      100%[=====] 351.55M  4.12MB/s   in 1m 45s
2024-12-12 14:35:54 (3.36 MB/s) - 'karaf-0.8.4.zip' saved [368625376/368625376]
dsbd@OLD-controler:~$
```

Décompresser le fichier:

```
dsbd@dsbd-virtual-machine: ~/karaf-0.8.4/bin
dsbd@dsbd-virtual-machine:~$ sudo apt install tar
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tar is already the newest version (1.34+dfsg-1ubuntu0.1.22.04.2).
tar set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 133 not upgraded.
dsbd@dsbd-virtual-machine:~$ sudo tar -xvf karaf-0.8.4.tar.gz
dsbd@dsbd-virtual-machine:~$ ls
Desktop    Downloads  karaf-0.8.4.tar.gz  Music      Public  Templates
Documents  karaf-0.8.4  karaf-0.8.4.tar.gz.1  Pictures   snap    Videos
dsbd@dsbd-virtual-machine:~$ cd karaf-0.8.4/
dsbd@dsbd-virtual-machine:~/karaf-0.8.4$ ls
bin          configuration  data      etc      LICENSE  system
build.url    CONTRIBUTING.markdown  deploy   lib      README.markdown  taglist.log
```

Aller dans le repertoire karaf-0.8.4/bin et exécutez la commande suivante pour lancer le contrôleur SDN :

```
sudo ./karaf
```

6. Installer Mininet

Installer mininet :

Installation de mininet : `sudo apt install mininet -y`

```

VMware-Virtual-Platform:~$ sudo apt install mininet -y
The following packages were automatically installed and are no longer required:
  linux-headers-6.11.0-8  linux-headers-6.11.0-8-generic  linux-modules-6.11.0-8-generic  linux-modules-extra-6.11.0-8-generic  linux-tools-6.11.0-8  linux-tools-6.11.0-8-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  mininet

Installing dependencies:
  cgroup-tools  libcgrou3  libunbound8  openvswitch-common  python3-openvswitch  python3-sortedcontainers
  iperf  libevent-2.1-7t64  libxdp1  openvswitch-switch  python3-packaging  socat

Suggested packages:
  openvswitch-doc  python-sortedcontainers-doc

Summary:
  Upgrading: 0, Installing: 13, Removing: 0, Not Upgrading: 44
  Download size: 5,254 kB
  Space needed: 17.6 MB / 18.5 GB available

Get:1 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-10 [144 kB]
Get:2 http://ma.archive.ubuntu.com/ubuntu oracular-updates/main amd64 libunbound8 amd64 1.20.0-1ubuntu2.1 [447 kB]
Get:3 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 openvswitch-common amd64 3.4.0-1 [1,057 kB]
Get:4 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 python3-sortedcontainers all 2.4.0-2 [27.6 kB]
Get:5 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 libxdp1 amd64 1.4.2-1ubuntu4 [62.5 kB]
Get:6 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 python3-openvswitch amd64 3.4.0-1 [1,036 kB]
Get:7 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 openvswitch-switch amd64 3.4.0-1 [1,632 kB]
Get:8 http://ma.archive.ubuntu.com/ubuntu oracular/universe amd64 libcgrou3 amd64 3.1.0-2 [62.9 kB]
Get:9 http://ma.archive.ubuntu.com/ubuntu oracular/universe amd64 cgroup-tools amd64 3.1.0-2 [85.0 kB]
Get:10 http://ma.archive.ubuntu.com/ubuntu oracular/universe amd64 iperf amd64 2.2.0-2dfsg-1 [154 kB]
Get:11 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 python3-packaging all 24.1-1 [41.4 kB]
Get:12 http://ma.archive.ubuntu.com/ubuntu oracular/main amd64 socat amd64 1.8.0-4build3 [374 kB]
Get:13 http://ma.archive.ubuntu.com/ubuntu oracular/universe amd64 mininet amd64 2.3.0-1.1 [131 kB]

```

A) Créez des topologies de réseau à l'aide de Mininet.

Voici un exemple de topologie linéaire avec 3 commutateurs et contrôleurs.

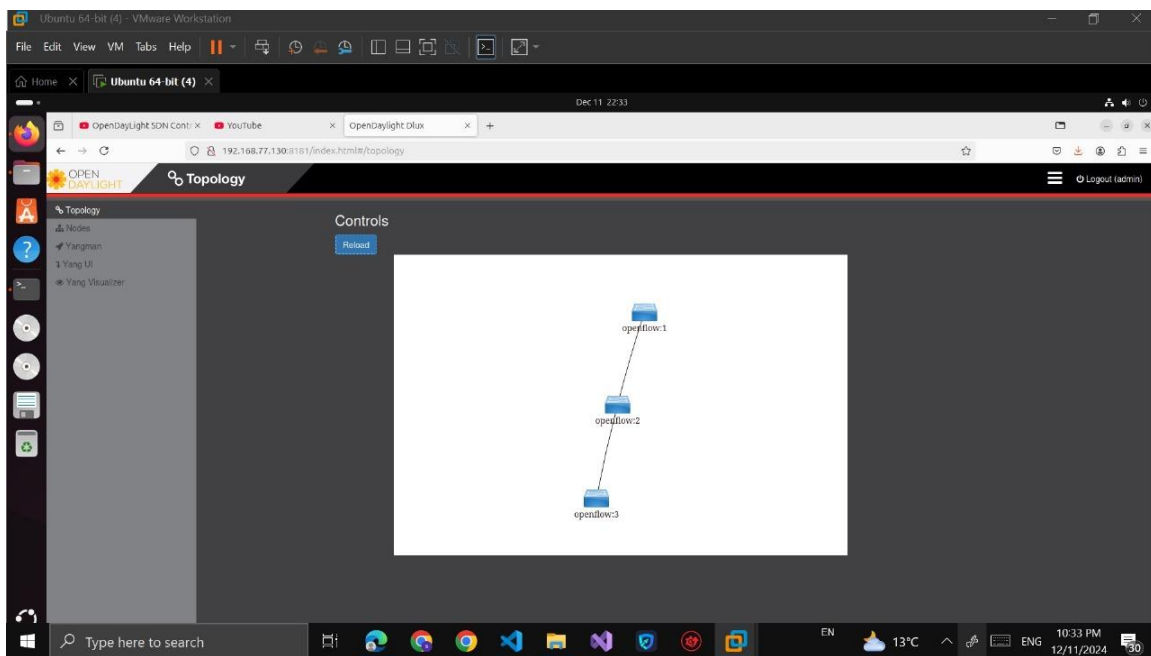
```

sudo mn --controller=remote,ip=Controller-IP --topo=linear,3
switch=ovsk,protocols=OpenFlow13

```

```
dsbd@OLD-controller: ~/mininet
dsbd@OLD-controller:~/mininet$ sudo mn --controller=remote,ip=192.168.78.146 --topo=linear,3 --switch ovsk,pr
otocols=OpenFlow13
*** Creating network
*** Adding controller
Unable to contact the remote controller at 192.168.78.146:6653
Unable to contact the remote controller at 192.168.78.146:6633
Setting remote controller to 192.168.78.146:6653
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet>
```

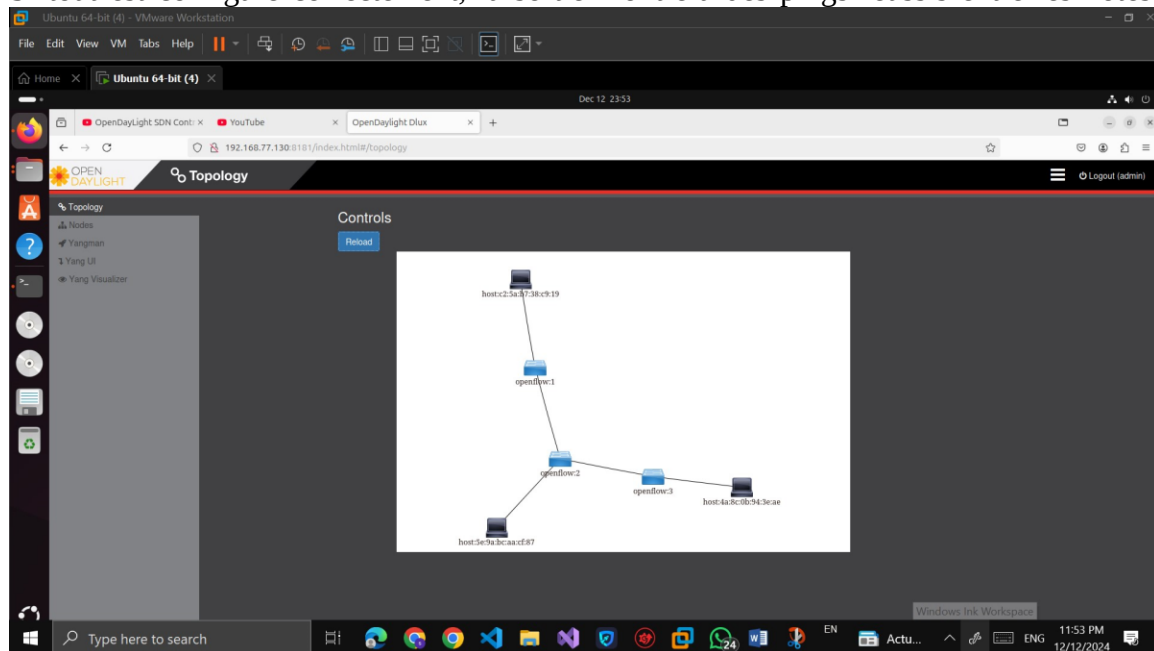
Aller sur le navigateur :



Testez la connectivité entre les hôtes de votre topologie. Commands:

```
mininet> pingall
```

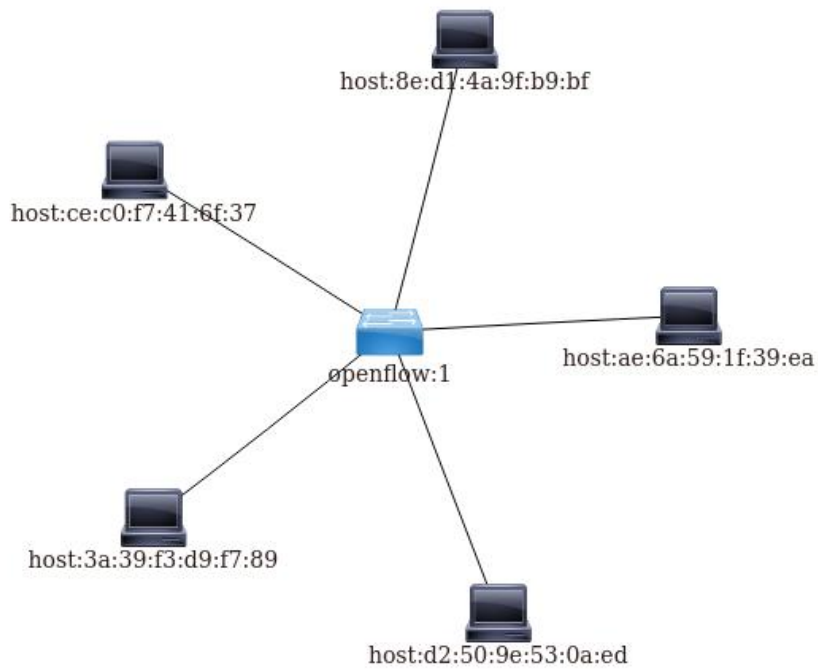
Si tout est configuré correctement, la sortie montrera des pings réussis entre les hôtes.



B) 2^{ème} Topologie

Utiliser exit pour sortir de mininet et créer la 2^{ème} topologie :

```
mininet@mininet-vm:~$  
mininet@mininet-vm:~$ sudo mn --controller=remote,ip=192.168.78.145 --topo=single,5 --switch=ovsk,pr  
otocols=OpenFlow13  
*** Creating network  
*** Adding controller  
Connecting to remote controller at 192.168.78.145:6653  
*** Adding hosts:  
h1 h2 h3 h4 h5  
*** Adding switches:  
s1  
*** Adding links:  
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1)  
*** Configuring hosts  
h1 h2 h3 h4 h5  
*** Starting controller  
c0  
*** Starting 1 switches  
s1 ...  
*** Starting CLI:  
mininet> pingall  
*** Ping: testing ping reachability  
h1 -> h2 h3 h4 h5  
h2 -> h1 h3 h4 h5  
h3 -> h1 h2 h4 h5  
h4 -> h1 h2 h3 h5  
h5 -> h1 h2 h3 h4  
*** Results: 0% dropped (20/20 received)  
mininet>
```

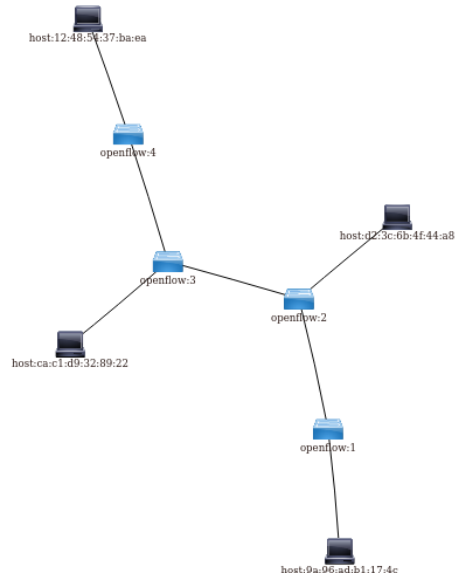


C) Créer une 3^{ème} topologie

```

Completed in 0:01:15.86 seconds
mininet@mininet-vm:~$ sudo mn --controller=remote,ip=192.168.78.145 --topo=linear,4 --switch=ovsk,protocols=OpenFlow13
*** Creating network
*** Adding controller
Connecting to remote controller at 192.168.78.145:6653
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3 s4
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (h4, s4) (s2, s1) (s3, s2) (s4, s3)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 4 switches
s1 s2 s3 s4 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> _

```



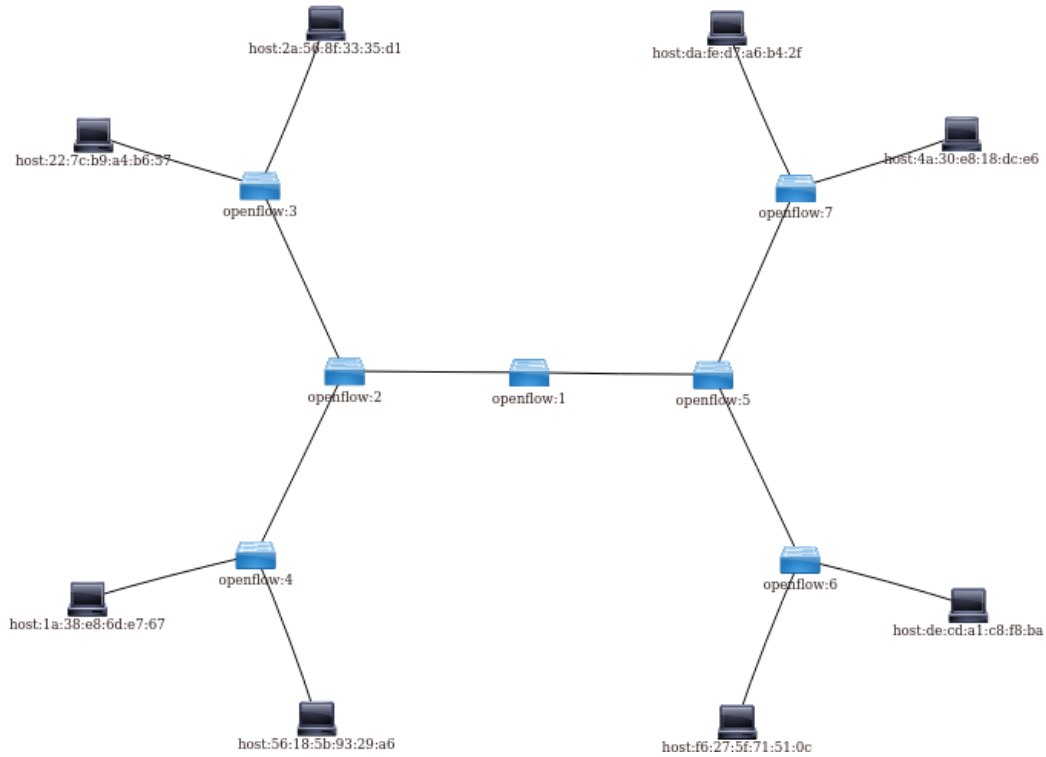
D)Créer une 4^{ème} topologie :

sudo mn --controller=remote,ip=Your_IP,topo=tree,depth=3,fanout=2
switch=ovsz,protocols=OpenFlow13

```

mininet@mininet-vm:~$ sudo mn --controller=remote,ip=192.168.78.145 --topo=tree,depth=3,fanout=2 --s
witch=ovsk,protocols=OpenFlow13
*** Creating network
*** Adding controller
Connecting to remote controller at 192.168.78.145:6653
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7
*** Adding links:
(s1, s2) (s1, s5) (s2, s3) (s2, s4) (s3, h1) (s3, h2) (s4, h3) (s4, h4) (s5, s6) (s5, s7) (s6, h5) (
s6, h6) (s7, h7) (s7, h8)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Starting controller
c0
*** Starting 7 switches
s1 s2 s3 s4 s5 s6 s7 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
h7 -> h1 h2 h3 h4 h5 h6 h8
h8 -> h1 h2 h3 h4 h5 h6 h7
*** Results: 0% dropped (56/56 received)
mininet>

```



E) lister les hôtes

La cli de Mininet nous permet de lister les éléments provisionnés (ou nodes) :

```
mininet> nodes
available nodes are:
c0 h1 h2 h3 h4 h5 h6 h7 h8 s1 s2 s3 s4 s5 s6 s7
mininet> _
```

F) Afficher les caractéristiques de chaque Hôte

Nous pouvons exécuter des commandes sur chaque noeud. Regardons les interfaces réseau disponibles sur chaque host :

```
mininet> h1 ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: h1-eth0@if646: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default
    qlen 1000
    link/ether 2a:56:8f:33:35:d1 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1/8 brd 10.255.255.255 scope global h1-eth0
        valid_lft forever preferred_lft forever
mininet> _
```

G) vérifier la connectivité entre hôtes

Mininet facilite la communication entre les différents noeuds. Exemple, nous pouvons pinger directement h1 depuis h3 de cette façon :

```

mininet>
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=3.30 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=1.79 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=1.98 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=1.81 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=2.30 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=1.57 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=1.61 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=1.49 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=1.66 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=1.71 ms
64 bytes from 10.0.0.3: icmp_seq=11 ttl=64 time=1.42 ms
^C
--- 10.0.0.3 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10717ms
rtt min/avg/max/mdev = 1.420/1.879/3.301/0.508 ms
mininet> _

```

8. Problèmes connus

Si vous rencontrez des problèmes, envisagez les solutions suivantes :

- OpenDaylight or Mininet is not starting properly (redémarrer opendaylight et installer les features).
- Check Java installation : java -version (il faut impérativement la version 8)
- Check OpenDaylight logs located in the data/log/karaf.log file for any errors.
- Mininet is not connecting to OpenDaylight.

Ensure the correct IP address and port are specified for the controller.